

Using the Tree to Improve the Sharpe Ratio

1.29

mean/ σ ratio, prob_included-weighted + tree tilt · vs.
1.25 without it

Rather than replacing the `prob_included` sizing rule, the tree's nested (leakage-free) predicted bucket from Slide 1 becomes a small multiplicative **tilt** on top of it — a modest adjustment sized to what 44% nested accuracy actually earns, not a wholesale re-sizing.

THE NEW POSITION-SIZING FORMULA

$$w_i = \left(\text{prob_included}_i / \text{mean}(\text{prob_included}) \right) \times \text{tilt}(\hat{b}_i)$$

w_i	final position weight for event i (equal-weight book = 1.0× average)
linear term	Week 5's sizing rule — weight proportional to child's index-inclusion probability
$\hat{b}_i$	tree's nested-LOO predicted bucket for event i (underperform / middle / outperform)

UNDERPERFORM

×0.7

MIDDLE

×1.0

OUTPERFORM

×1.3

MEAN / Σ OF 252D HEDGED CAR

Equal-weight		0.69
prob_included-wt.		1.25
+ tree tilt		1.29

MEAN CAR, TILTED

+23.5%

 Σ , TILTED

18.2%

Each event's tilt comes from a tree that never saw that event's own label during alpha selection or fitting — the same discipline used to score the tree's 44% accuracy on Slide 1.

n=16

Read as hypothesis, not strategy. The tilt moves the ratio from ~1.25 to ~1.29 — real and leakage-free, but small, and built on a tree whose own nested accuracy barely clears the majority baseline. The **prob_included-weighted book alone (1.25)** has the stronger evidence; the **tree tilt (1.29)** is a documented next lead to keep testing as more spinoffs complete, not a validated upgrade.